Research on Clearness Index Support Vector Regression Method for Solar Radiation Forecasting

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***Abstract*—Aiming at the problem of intermittent and fluctuating solar radiation, which is difficult to forecast accurately, a short-term forecasting method of solar radiation intensity based on support vector regression is proposed. First, the trend curve of the daily clearness index is calculated by using the K-means algorithm to cluster the curves to obtain the cluster center of the curve under the general weather type. Then, the gradient boosting decision tree is used to judge the forecasting weather type, and the corresponding weather type is used. The meteorological information builds a support vector regression model from time to time to obtain the clearness index and calculate the solar radiation intensity, which avoids the problem of low forecasting accuracy caused by using the same fitting model in different weather conditions. Finally, the proposed forecasting method is verified with measured data. Experimental results show that the proposed method has high forecasting accuracy.**

Keywords—solar radiation intensity forecasting; support vector regression; clearness index; K-means; gradient boosting decision tree

I. INTRODUCTION

With the increase in the grid connected capacity of photovoltaic power stations, the uncertainty of photovoltaic power generation brings severe challenges to power system dispatching and operation[1,2]. Photovoltaic power generation is mainly affected by solar radiation intensity. The establishment of an accurate short-term forecasting model for solar radiation intensity is of great significance for the power grid dispatching department to formulate the day ahead dispatching plan[3-5].

At present, there are physical forecasting methods, cloud map-based forecasting methods and data-based statistical forecasting methods for solar radiation intensity forecasting. The physical forecasting method calculates the solar radiation intensity in the clearness by constructing the solar radiation transmission model[6]. Reference [7] used the simplified REST2 model to realize the rolling prediction of solar radiation intensity clearness by predicting the atmospheric aerosol optical thickness (AOD). The cloud map prediction method used the cloud map to analyze and model with amount and cluster velocity of clouds. Due to the influence of the sky imager view angle and wind speed, its time scale was short, so it cannot be applied to the prediction and modeling of solar radiation sequences[8,9]. Statistical prediction methods use machine learning methods to establish the mapping relationship between historical data or numerical weather forecasts and future values to obtain the prediction results. Reference [10] established a nonlinear auto-regressive network forecasting model by selecting the optimal meteorological parameters and achieved good forecasting results. Reference [11] established support vector regression and artificial neural network photovoltaic power forecasting models, and its experimental results showed that the support vector regression model had higher forecasting accuracy and less computational complexity. Reference [12] used an optimization algorithm to optimize the super-parameters of a support vector regression forecasting model, which further improved the forecasting accuracy.

The prediction accuracy of the SVR model would be reduced by applying it to the generalization fitting of the same model for different weather types when sudden change weather is encountered. Cluster analysis can be used to analyze and process different weather types[13]. However, the solar radiation intensity at each time in summer is higher than that at the corresponding time in winter. The number of clusters would be too large, and the model learning effect would be bad if we clustered the solar radiation intensity change curves directly according to weather type. To avoid the impact of periodic changes, the clearness forecasting index could be obtained by solar radiation intensity. However, the clearness index calculation needed to know the AOD, water vapor content and other parameters of the forecasting place[14]. It is difficult for AOD parameters to obtain in real time[15]. Relatively, it is simpler to calculate the clearness index through extraterrestrial solar radiation intensity, which avoids the calculation interference of various components in the atmosphere [16]. Experiments in reference [17] showed that using the clearness index to forecast solar radiation intensity could achieve higher accuracy than direct forecasting.

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The method of using the clearness index to realize weather category clustering is proposed in this paper, which uses the corresponding historical weather type data to train the support vector regression model to improve the forecasting accuracy. First, the weather type was divided by clustering the change trend curve of the daily clearness index. The classification rules of the gradient boosting decision tree were established by using weather type labels obtained by historical weather information, change trend curves of the clearness index and clustering. The weather type of the forecasting day was obtained by using classification rules, historical clearness index and meteorological information. Then, the support vector regression model is established hourly. The model was trained using historical data corresponding to time under the same weather type. Finally, the daily meteorological information was used as the model input to obtain the clearness index at each time and to calculate the solar radiation intensity. The effectiveness of the proposed method was verified by a case analysis.

II. Calculation of Extraterrestrial Solar Radiation and Clearness Index

When solar radiation is transmitted from the outside atmosphere to the ground, some components, such as aerosols, water vapor and dust, weaken solar radiation. At the same time, due to the Earth's revolution and periodic atmospheric phenomena, the solar radiation intensity presented an annual periodic variation law. The change curve of the average solar radiation intensity of a place from 10:00 to 11:00 every day in 2015 and 2016 is shown in Figure 1. In order to reduce the seasonal influence, the clearness index was forecasted instead of the direct forecasting of solar radiation to improve the forecasting accuracy.

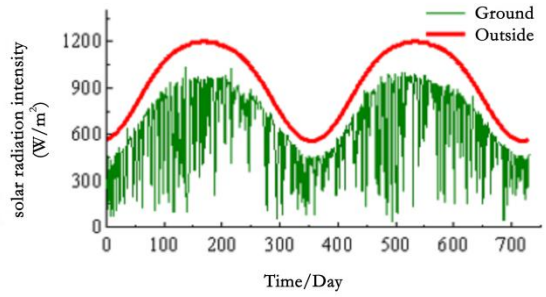


Fig. 1. Changes in the average solar radiation intensity between 10 and 11 a.m. in a place in 2015 and 2016

The clearness index K_t is the ratio of the solar radiation G_t received by the Earth's horizontal plane to the solar radiation $G_{0,t}$ received by the extra atmospheric water plane in a period of time t:

$$k = G_t / G_{0,t} \tag{1}$$

The extraterrestrial solar radiation intensity [18] per unit area of the horizontal plane at any place and time is expressed in formula (2).

$$G = G_{sc} k(\cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta) \tag{2}$$

In the above formula, G_{sc} is the solar constant, with a value of 1368 W/m^2; k is the correction coefficient of solar terrestrial distance; φ is the location latitude; δ is the solar declination angle; and ω is the solar hour angle. The total solar radiation in one hour was calculated. When we hypothesized ω_h was the solar time angle at the middle time of any hour, combined with the Earth rotation angle velocity, the total radiation value within one hour could be calculated using formula (3).

$$G_{0,t} = \frac{43200}{\pi} G_{sc} k \int_{\omega_h-\pi/12}^{\omega_h+\pi/12} (\cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta) d\omega \tag{3}$$

At the same time, the total extraterrestrial solar radiation [10] per unit area of horizontal plane at any place within one day was calculated in formula (4).

$$G_{0,d} = \frac{24}{\pi} G_{sc} k(\omega_{sr}\sin\delta\sin\varphi + \cos\delta\cos\varphi\sin\omega_{sr}) \tag{4}$$

In equation (4), ω_{sr} was sunrise azimuth.

The clearness index can reflect the influence of various components in the atmosphere on solar radiation transmission. The change curve of the clearness index was used to realize the division of weather types, and the forecasting model was trained according to weather types to improve the forecasting accuracy.

III. Prepare Your Paper Before Styling

A. Daily Clearness Index Change Trend Curve

According to the calculation formula in the previous section, the clearness index of any place and any time period can be calculated. The solid line in Figure 2 shows the change in the hourly clearness index curve over two days at a solar radiation observation station. It can be seen from the figure that the clearness index had the same change trend. In order to eliminate different cases of the same clearness index about curve change and realize curve clustering under the same weather type. By subtracting the daily clearness index k_d of the day from the clearness index $k_{h,d}$ at each time, the change trend curve $k^*_{h,d}$ of the daily clearness index can be obtained.

$$k^*_{h,d} = k_{h,d} - k_d \tag{5}$$

It can be seen from the dotted line after the change in Figure 2 that the values of the two-day clearness index curve are roughly the same to realize the clustering of curves of the same weather type.

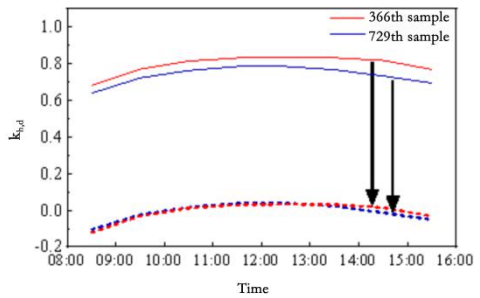


Fig. 2. Two days with similar changes in clearness index curve

B. Weather Type Clustering and Judgment Based on K-means and Gradient Boosting Decision Trees

Cluster analysis can classify unknown experimental data with the same or similar characteristics into the same

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category by using the similarity measurement method to realize classification. As a dynamic clustering algorithm, *K*-means has the characteristics of simple principles and fast convergence speed. It was efficient for large data sets.

The obtained trend curves of the daily clearness index were clustered to determine the approximate change trend of the daily clearness index under the same weather type. Using the *K*-means clustering method, the weather state was divided into four generalized weather types A, B, C and D, according to the change trend of the clearness index, and the clustering centers of the four change trend curves were calculated. Its calculation method was as follows,

(1) In the sample data, four clearness index change trend curves were randomly selected as the clustering center;

(2) Calculate the Euclidean distance from other change trend curves to each cluster center curve, and select the minimum distance as the clustering category of the change trend curve of the clearness index;

(3) Calculate the average value of each point of all clearness index change trend curves of each category, and its connecting line was used as the new curve clustering center of the category;

(4) Steps (2) ~ (3) were repeated until the cluster center of the clearness index change trend curve did not change.

After clustering weather types, it was necessary to judge the weather type of the forecast day, realized by a gradient boosting decision tree multi-classification algorithm.

The gradient boosting decision tree (GBDT) is an ensemble learning algorithm with strong generalization performance. When applying GBDT to the regression problem, the training data were set as $T=\{(x_i, y_i), i=1,2,\ldots,N\}$, the number of iterations was M, and the loss function was recorded as $L(y, f(x))$.

Initialize the weak classifier firstly,

$$f_0(x)=\arg\min_c \sum_{i=1}^{M} L(y_i, c) \tag{6}$$

The negative gradient was calculated for each training sample at each iteration. When the number of iterations was m, the negative gradient of sample i can be expressed in formula (7).

$$r_{mi}=-\left[\frac{\partial L(y_i, f(x_i))}{\partial(x_i)}\right]_{f(x)=f_{m-1}(x)} \tag{7}$$

At the same time, the tree region with leaf nodes was fitted. The number of leaf nodes was J, and the node domain was R_{mj}, where $j=1,2,\cdots, J$. The best fitting value of the node domain can be calculated by formula (8).

$$c_{mj}=\arg\min_c \sum_{x_i\in R_{mj}} L(y_i, f_{m-1}(x_i)+c) \tag{8}$$

Then, update the learner.

$$f_m(x)=f_{m-1}(x)+\sum_{j=1}^{J} c_{mj} I(x\in R_{mj}) \tag{9}$$

Finally, a strong learner was obtained.

$$f(x)=f_0(x)+\sum_{m=1}^{M}\sum_{j=1}^{J} c_{ij} I(x\in R_{mj}) \tag{10}$$

In this paper, GBDT was applied to four generalized types of weather classification problems, and the logarithmic loss function was selected.

$$L(y, f(x))=-\sum_{k=1}^{K} y_k \log p_k(x) \tag{11}$$

Considering that the number of output categories T was k and $y_k=1$, the k-th sample category probability $p_k(x)$ was expressed as formula (12).

$$p_k(x)=\exp(f_k(x))\Big/\sum_{l=1}^{k}\exp(f_l(x)) \tag{12}$$

Then, the negative gradient error of the loss function corresponding to the i-th sample category L in the m-th iteration can be expressed in formula (13).

$$r_{mil}=y_{il}-p_{l,m-1}(x_i) \tag{13}$$

The GBDT model was established to judge the current weather type. The meteorological information of the forecast day was used as the input of the model. The forecasting time length was 8 hours in this paper. In order to simplify the input of the model, the meteorological information selected three average values of temperature, humidity, cloud cover and air pressure, daily average value, 8-12 hour average value and 12-15 hour average value. At the same time, we took the whole day clearness index two days before the forecast day as the input of the model and took the label of generalized weather type as the model output to judge the change trend curve of the current clearness index.

C. Support Vector Regression Forecasting Model of The Clearness Index

The core of the support vector machine (SVM) was the VC dimension theory of statistical learning and the principle of structural risk minimization. On this basis, SVM was applied to regression analysis by introducing an insensitive loss function ε. It can solve practical problems successfully, such as nonlinearity, high dimensionality and local extrema, and its fitting performance is better than that of the traditional artificial neural network algorithm.

The purpose of support vector regression (SVR) analysis was to find the minimum total deviation of all sample points from the hyperplane. Introduce the insensitive loss function ε, which was defined as loss, especially when the error between the fitting value and the actual value exceeds the parameter ε. This method improves the generalization and robustness performance of the model.

The regression function can be expressed as follows.

$$f(x)=\omega^T\Phi(x)+b \tag{14}$$

Introducing non-negative relaxation variables ξ_i and ξ_i^*, solving problems can be summarized as formula (15).

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$$\min\ Q=\frac{1}{2}\|\omega\|^2+C\sum_{i=1}^{N}(\xi_i+\xi_i^*)$$
$$s.t.\begin{cases} y_i-\omega^T\Phi(x_i)-b\le\varepsilon+\xi_i \\ \omega^T\Phi(x_i)+b-y_i\le\varepsilon+\xi_i^* \\ \xi_i\ge 0,\xi_i^*\ge 0,i=1,2,\cdots,N \end{cases} \quad (15)$$

When the training data were linearly separable, the original input data were mapped to the high-dimensional feature space to make the data linearly separable. In this paper, the radial basis function is selected as the kernel function in formula (16).

$$K(x_i,x_j)=\exp(-\|x_i-x_j\|^2/\sigma^2) \quad (16)$$

For the constraints in equation (15), two non-negative multipliers are introduced α_i and α_i^*, and the regression function can be calculated according to the Lagrange algorithm in formula (17).

$$f(x)=\sum_{i=1}^{N}(\alpha_i-\alpha_i^*)K(x_i,x)+b \quad (17)$$

A support vector regression model was established at each time point. The input of the model was temperature, humidity, cloud cover and air pressure at this time under the same weather type in the historical data. Output was the clearness index k_d corresponding to this time. The network structure diagram of the model is shown in Figure 3.

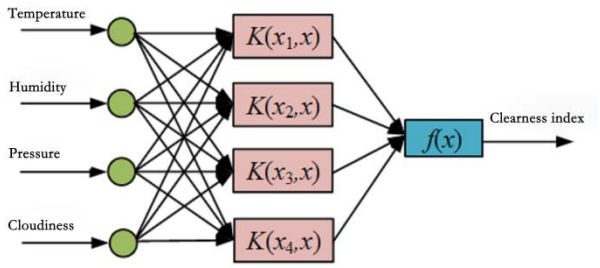


Fig. 3. Support vector regression structure diagram

D. Short-term Forecasting of The Overall Process of Solar Radiation Intensity

Considering the steps discussed above, the variation trend curve of the daily clearness index was calculated according to the historical solar radiation intensity and the data of forecasting site geographical location. K-means was used to obtain the curve clustering center in the case of generalized weather types. GBDT was used to judge the daily weather type forecasting, and an SVR model was established hourly. The model was trained with the same generalized weather type data. The forecasting daily meteorological data were used as the forecasting input. Therefore, the clearness index at each time can be obtained, and the solar radiation intensity can be calculated. The overall flow chart is shown in Figure 4.

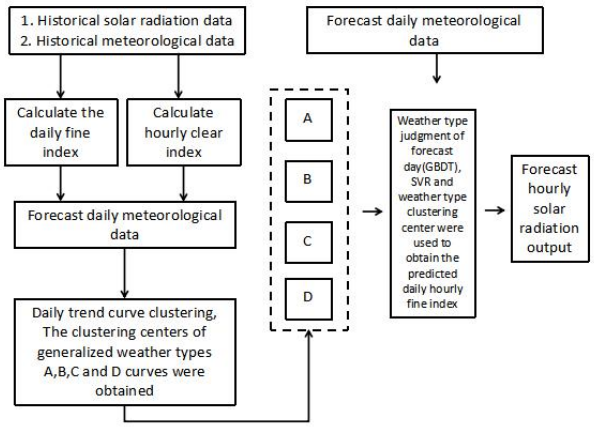


Fig. 4. Overall process of short-term solar radiation intensity forecasting

IV. Simulation Experiment

A. Data Sources

In order to verify the forecasting effect of the model, the observation data of the American renewable energy laboratory from 2015 to 2017 were used for simulation experiments. The data include the average value of observation data per hour and mainly include 5 indexes: temperature, humidity, cloud cover, air pressure and solar radiation intensity. The outdoor solar radiation intensity in each hour was calculated from 8:00 to 15:00 on the corresponding date. The first 90% of the data were selected as the training dataset, and the last 10% of the data were used as the test dataset to verify the forecasting performance of the proposed method.

B. Forecasting Performance Indicators

This paper selects solar radiation intensity forecasting evaluation indexes, commonly used in the field, relative root mean square error (*rRMSE*) and s relative mean absolute error (*rMAE*). Their calculation method can be expressed in formula (18) and formula (19).

$$rRMSE=\sqrt{\frac{1}{N}\sum_{i=1}^{N}(G_{Pi}-G_{Mi})^2}\Big/\bar{G}_{Mi} \quad (18)$$

$$rMAE=\frac{1}{N}\sum_{i=1}^{N}\left|G_{Pi}-G_{Mi}\right|\Big/\bar{G}_{Mi} \quad (19)$$

In the above formulas, G_{Pi} represents the forecasting val ue of solar radiation intensity; G_{Mi} represents the actual mea sured value of solar radiation intensity; $\bar{G}_P$ represents the av erage of the forecasting value series; $\bar{G}_M$ represents the aver age value of the actually measured solar radiation value seq uence; and N indicates the number of data collection points.

C. Clustering Results of Training Dataset

The variation trend curve of the daily clearness index of the training data set was calculated. The curves were cluster ed with K-means clustering to obtain the clustering center c urve under four generalized weather types, which are shown in Fig. 5.

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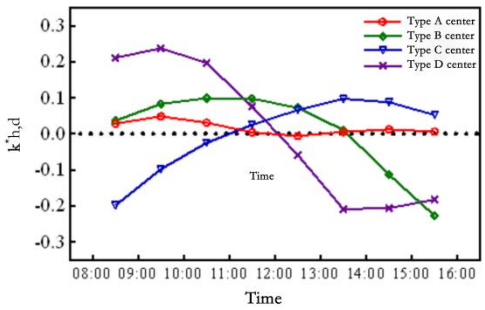


Fig. 5. Clustering center of clearness index change trend curve

Tab. 1 shows the days and percentages corresponding to each weather type in the training data set. The correspondin g days of weather type A were the highest, accounting for 3 5.06%, and the proportion of weather type D was the lowest.

TABLE I. NUMBER AND PERCENTAGE OF EACH WEATHER TYPE IN THE TRAINING DATA

Type	A	B	C	D
Days	346	286	203	152
Proportion	35.06%	28.97%	20.57%	15.40%

D. Forecasting Results and Analysis

For the 109-day test dataset, the model established in thi s paper was used for forecasting. The forecasting performan ce indexes of each generalized weather type in the test data s et are calculated, as shown in Table 2. With the four general ized weather types A, B, C and D, one day was selected for t he comparison between the forecasting value and, , as show n in Figures 6 to 9. The results indicated that each time point achieved high forecasting accuracy under various weather t ypes with various weather types. Among them, weather type D accounted for the least days in the test data set (14 days i n total), and the forecasting accuracy obtained by the model was low. This showed that unstable weather would affect th e forecasting accuracy of the model.

TABLE II. TEST DATASET FORECASTING PERFORMANCE

Type	*rRMSE*	*rMAE*
A	0.222	0.168
B	0.159	0.119
C	0.236	0.180
D	0.254	0.192

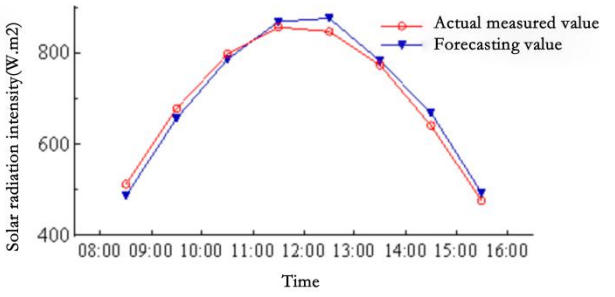


Fig. 6. Test dataset forecasting and actual value curve on day 5 (Type A)

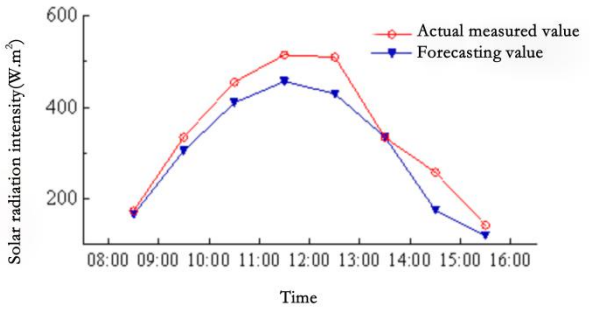


Fig. 7. Test dataset forecasting and actual value curve on day 102 (Type B)

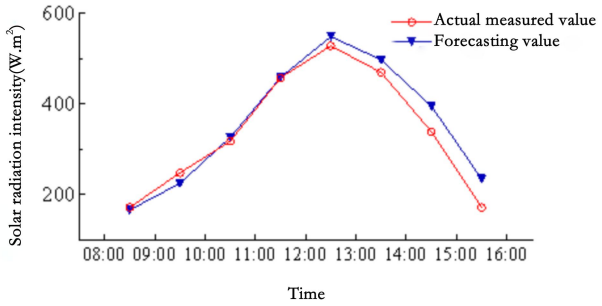


Fig. 8. Test dataset forecasting and actual value curve on day 68 (Type C)

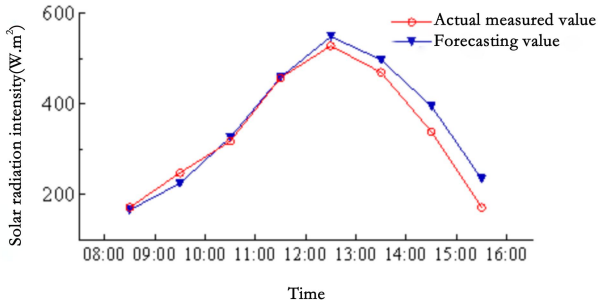


Fig. 9. Test dataset forecasting and actual value curve on day 9 (Type D)

In order to further verify the forecasting effect of the method proposed in this paper, the traditional BP neural network model and support vector regression model were used to predict the solar radiation intensity for the test dataset. The forecasting performance of all models in the test dataset is shown in Table 3. The test dataset was selected, and the comparison curve was generated on the 5th day. Figure 10 and Table 3 show that the forecasting performance of the proposed method was improved compared with that of the nonclustered SVR method, with *rRMSE* reduced by 0.071 and *rMAE* reduced by 0.051.

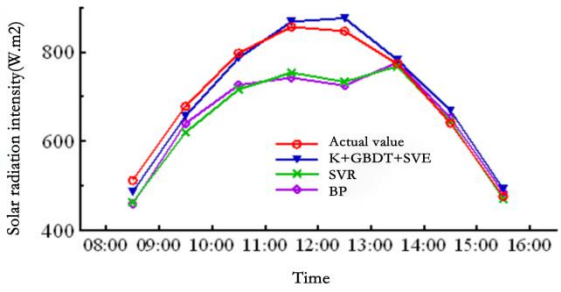


Fig. 10. Comparison curve of different forecasting methods on the 5th day of the test dataset

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TABLE III. COMPARISON OF PERFORMANCE INDICATORS OF DIFFERENT FORECASTING METHODS

Type	*rRMSE*	*rMAE*
K+GBDT+SVR	0.210	0.159
SVR	0.281	0.210
BP	0.292	0.220

V. CONCLUSION

Solar radiation intensity is the main factor affecting the power generation of photovoltaic power stations. In this paper, the *K*-means algorithm is used to cluster the change trend curve of the clearness index to realize the division of generalized weather types. We use the gradient boosting decision tree to judge the weather type of the forecast day and select the historical data under the same weather type to train the support vector regression model. The first mock exam is used to predict the daily weather information, and then the solar radiation intensity is calculated, which avoids poor forecasting results by using the same model in different weather types. The simulation results of the measured data show that the forecasting accuracy of the proposed method is higher than that of the traditional BP neural network model and the nonclustered SVR model.

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